

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
Student Name:	AKSHAYA KUMAR CHANDRASEKAR	Reg. No.:	192425350
Section / Batch:	B.TECH AIML	Date:	28-07-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

A mapping application determines the outer boundary of a set of plotted land coordinates using a brute-force convex hull method. The dataset size increases significantly over time.

- Explain how the brute-force convex hull algorithm works in this case.
- Analyze the computational complexity of checking all point pairs.
- Discuss why this approach becomes inefficient for large inputs.

main.py

Share

Run

Output

```

1 points = [(1,1), (2,5), (5,3), (3,2), (4,6)]
2
3 n = len(points)
4
5 print("Possible Convex Hull Edges")
6
7 for i in range(n):
8     for j in range(i + 1, n):
9         a = points[i]
10        b = points[j]
11
12        pos = 0
13        neg = 0
14
15        for k in range(n):
16            if k == i or k == j:
17                continue
18
19            c = points[k]
20
21            value = ((b[0]-a[0])*(c[1]-a[1]) -
22                    (b[1]-a[1])*(c[0]-a[0]))

```

Possible Convex Hull Edges

(1, 1) -- (2, 5)

(1, 1) -- (5, 3)

(1, 1) -- (3, 2)

(2, 5) -- (4, 6)

(5, 3) -- (3, 2)

(5, 3) -- (4, 6)

=== Code Execution Successful ===

Code of Conduct

I AKSHAYA KUMAR CHANDRASEKAR(192425350) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Akshaya kumar

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding ; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex real world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision Making Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions

Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____